International Islamic University Chittagong

Department of Science of Hadith and Islamic Studies

Microcomputer Application II

Course Lecture Notes

Course Information

- **Course Code:** CSE-3503
- **Course Title:** Microcomputer Application II
- **Credit Hours:** 3 (Three)
- **Contact Hours:** 3 (Three)
- **Program:** B.A. (Hon's) for all departments of SHIS

Instructor Information

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Academic Year: 2026

Marks Distribution

The total marks for the course are **100**.

Assessment Component	Marks
Attendance	10
Assignment / Class Test	10
Mid-Term Examination	30
Final Examination	50
Total	**100**

Lecture 01

Microsoft Access: Database Basics, Fields, Tables and Keys

1. Introduction to Data

Before understanding a database, we first need to understand the term **data**.

Definition: Data

Data refers to raw facts, observations, numbers, text, symbols or values that can be stored and processed by a computer.

Examples of data include:

- Student ID: 2026001
- Student Name: Abdullah
- Department: SHIS
- Semester: 4
- CGPA: 3.75
- Mobile Number: 018XXXXXXXX

Individual pieces of data may not provide much meaning. When data are organized and processed, they become useful **information**.

Example

Consider the following values:

$$2026001, \quad Rahim, \quad SHIS, \quad 3.85$$

Individually, these are data.
When we say:

Student ID 2026001 belongs to Rahim from SHIS and his CGPA is 3.85.

the organized data become meaningful information.

2. What is a Database?

Definition: Database

A database is an organized collection of related data that can be stored, accessed, managed and updated efficiently.

For example, a university may maintain a database containing:

- Student information
- Teacher information
- Course information
- Examination results
- Attendance
- Library records
- Tuition payment information

Instead of keeping all information in separate paper files, the information can be stored systematically inside a database.

3. Traditional File System vs Database

Suppose a department maintains student information in several separate Excel files or paper documents.

The same student's name, ID and phone number may appear repeatedly.

This may cause:

- Duplicate information
- Difficult searching
- Data inconsistency
- Difficult updating
- Poor security
- Higher chance of errors

A database solves many of these problems by storing data in an organized form.

Traditional File System	Database System
Data may be duplicated	Reduces unnecessary duplication
Difficult to search	Easier to search
Difficult to update	Easier to update
More inconsistency	Better consistency
Limited relationships	Supports relationships between data
Manual management	Structured data management

4. Database Management System

Definition: DBMS

A Database Management System (DBMS) is software used to create, store, organize, retrieve, update and manage data in a database.

Common examples of DBMS software include:

- Microsoft Access
- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite

5. Introduction to Microsoft Access

Microsoft Access is a relational database management system developed by Microsoft.

It is included with some editions of Microsoft Office and Microsoft 365.

Microsoft Access allows users to create and manage databases without requiring advanced programming knowledge.

It is particularly suitable for:

- Small business databases
- Student databases
- Library management
- Employee databases
- Inventory systems
- Customer records
- Academic record management

6. Main Objects in Microsoft Access

Microsoft Access contains several important database objects.

1. **Tables**

   Tables are used to store data.

2. **Queries**

   Queries are used to search, filter, calculate and retrieve specific data.

3. **Forms**

   Forms provide a user-friendly interface for entering and viewing data.

4. **Reports**

   Reports are used to display and print information in an organized format.

A simple way of remembering them is:

$$\boxed{\text{Table} \rightarrow \text{Query} \rightarrow \text{Form} \rightarrow \text{Report}}$$

7. Database Table

Definition: Table

A table is a database object that stores related data in rows and columns.

A database normally contains one or more tables.

For example, a university database may contain:

- Student table
- Teacher table
- Course table
- Department table
- Result table

Consider the following **Student** table.

StudentID	Name	Department	CGPA
101	Rahim	SHIS	3.80
102	Karim	ELL	3.65
103	Abdullah	QSIS	3.91
104	Hasan	SHIS	3.55

Here:

- The complete structure is a **table**.
- Each column represents a **field**.
- Each row represents a **record**.

8. Field

Definition: Field

A field is a single category or attribute of information stored in a table. Fields normally appear as columns in a database table.

For example, the Student table may contain the following fields:

- StudentID
- StudentName
- Department
- DateOfBirth
- MobileNumber
- Email
- CGPA

Consider:

StudentID	StudentName	Department	CGPA
101	Rahim	SHIS	3.80

Here, **StudentID**, **StudentName**, **Department** and **CGPA** are fields.

9. Record

Definition: Record

A record is a complete set of related fields describing one person, object, event or item. Records normally appear as rows in a table.

For example:

101 Rahim SHIS 3.80

This complete row represents one **student record**.

Therefore:

$$\boxed{\text{Table} = \text{Collection of Records}}$$

and

$$\boxed{\text{Record} = \text{Collection of Related Fields}}$$

10. Field vs Record

Field	**Record**
Represents one attribute	Represents one complete entity
Usually displayed as a column	Usually displayed as a row
Example: StudentName	Example: information of Student 101
Contains one type of data	Contains several related fields

11. Field Data Types in Microsoft Access

When creating a field, we must specify what type of data the field will contain.

Some commonly used Microsoft Access data types are:

Data Type	Purpose
Short Text	Stores text, numbers that are not used for calculations, names, phone numbers and codes.
Long Text	Stores long paragraphs or descriptions.
Number	Stores numerical values used in calculations.
Date/Time	Stores dates and times.
Currency	Stores monetary values.
AutoNumber	Automatically generates a unique number for each record.
Yes/No	Stores Boolean values such as Yes/No or True/False.
Hyperlink	Stores web addresses or email links.
Attachment	Stores files such as images or documents.

Example Field Design

A student table could be designed as:

Field Name	Data Type	Purpose
StudentID	AutoNumber	Unique student number
StudentName	Short Text	Student name
Department	Short Text	Department name
DateOfBirth	Date/Time	Birth date
CGPA	Number	Academic result
ActiveStudent	Yes/No	Student status

12. Database Keys

Keys are extremely important in relational databases.

Definition: Key

A key is a field or combination of fields used to identify records and establish relationships between database tables.

The most important types of keys are:

1. Primary Key
2. Candidate Key
3. Alternate Key
4. Foreign Key
5. Composite Key

12.1 Primary Key

Primary Key

A primary key is a field or combination of fields that uniquely identifies each record in a table.

Example:

StudentID	Name	Department	CGPA
101	Rahim	SHIS	3.80
102	Karim	ELL	3.65
103	Hasan	SHIS	3.70

Here,

$$\boxed{\text{StudentID = Primary Key}}$$

because every StudentID is unique.

Characteristics of a Good Primary Key

A primary key should:

- Be unique.
- Never contain duplicate values.
- Normally not contain NULL values.
- Remain stable whenever possible.
- Identify exactly one record.

12.2 Candidate Key

A **candidate key** is any field that is capable of uniquely identifying a record.

Suppose a Student table contains:

- StudentID
- NationalID
- UniversityEmail
- StudentName

If StudentID, NationalID and UniversityEmail are all unique, then all three could potentially identify a student.

Therefore, they are candidate keys.

One of them is selected as the primary key.

12.3 Alternate Key

Candidate keys that are not selected as the primary key are called **alternate keys**.

For example:

- StudentID — Primary Key
- NationalID — Alternate Key
- UniversityEmail — Alternate Key

12.4 Foreign Key

Foreign Key

A foreign key is a field in one table that refers to the primary key of another table.

Consider two tables.

Department Table

DepartmentID	DepartmentName
D01	SHIS
D02	ELL
D03	QSIS

Here:

$$DepartmentID = Primary\ Key$$

Now consider:

Student Table

StudentID	Name	DepartmentID
101	Rahim	D01
102	Karim	D02
103	Hasan	D01

In the Student table:

$$DepartmentID = Foreign\ Key$$

because it refers to the DepartmentID of the Department table.

12.5 Composite Key

A **composite key** consists of two or more fields that together uniquely identify a record.

Consider an Enrollment table.

StudentID	CourseID	Semester
101	CSE3503	Spring 2026
101	ENG2301	Spring 2026
102	CSE3503	Spring 2026

StudentID alone is not unique.

CourseID alone is also not unique.

But the combination:

$$\boxed{StudentID + CourseID}$$

can uniquely identify an enrollment record.

This is called a **composite key**.

13. Summary of Database Keys

Key Type	**Description**
Primary Key	Uniquely identifies each record in a table.
Candidate Key	A field that can potentially become the primary key.
Alternate Key	A candidate key that is not selected as the primary key.
Foreign Key	A field that references the primary key of another table.
Composite Key	A key consisting of two or more fields.

14. Basic Table Relationships

Database tables are often connected using keys.

Consider:

$$\boxed{\text{Department}} \longrightarrow \boxed{\text{Student}}$$

One department may contain many students.

This is called a:

$$\boxed{\text{One-to-Many Relationship}}$$

For example:

Department D01

$\downarrow$

Student 101

Student 103

Student 107

Student 115

The DepartmentID appears once in the Department table but may appear many times in the Student table.

15. Practical: Creating a Database in Microsoft Access

Students can perform the following exercise during the class.

15.1 Step 1: Open Microsoft Access

Open:

Start Menu → Microsoft Access

15.2 Step 2: Create a Blank Database

Select:

Blank Database

Enter the database name:

`UniversityDB.accdb`

Then click:

Create

15.3 Step 3: Create a Student Table

Create the following fields:

Field Name	Data Type	Key
StudentID	AutoNumber	Primary Key
StudentName	Short Text	–
Department	Short Text	–
MobileNumber	Short Text	–
DateOfBirth	Date/Time	–
CGPA	Number	–

15.4 Step 4: Set the Primary Key

Select the **StudentID** field and click:

Table Design → Primary Key

A key symbol should appear beside StudentID.

15.5 Step 5: Save the Table

Save the table as:

`Student`

15.6 Step 6: Enter Sample Records

Enter the following data.

StudentID	StudentName	Department	CGPA
1	Rahim Ahmed	SHIS	3.70
2	Abdullah Khan	QSIS	3.80
3	Karim Hasan	ELL	3.45
4	Hasan Ali	SHIS	3.90
5	Mahmud Rahman	QSIS	3.65

16. Real-Life Example

Suppose IIUC wants to maintain information about students.

Instead of storing everything in one large table, the database could contain:

Students Courses Departments Teachers Results

The tables can then be linked together using primary and foreign keys.

For example:

$$\text{Department} \xrightarrow{\text{DepartmentID}} \text{Student}$$

and

$$\text{Student} \xrightarrow{\text{StudentID}} \text{Result}$$

This organization makes it easier to manage a large amount of information.

17. Common Mistakes

Students should avoid the following mistakes when creating tables:

1. Using student names as primary keys.

   Names may be duplicated.

2. Using Number data type for mobile numbers.

   Mobile numbers are identifiers, not values used for mathematical calculation. Therefore, **Short Text** is usually more appropriate.

3. Keeping multiple pieces of information in one field.

   Instead of:

   `Rahim, SHIS, 018XXXXXXXX`

   use separate fields:

   StudentName Department MobileNumber

4. Allowing duplicate values in a primary key.

5. Selecting an inappropriate data type.

18. Class Activity

Design a database table named **Book** for a university library.

The table should contain at least the following information:

- Book ID
- Book Title
- Author
- Publisher
- Publication Year
- Price
- Available or Not

Students should:

1. Decide suitable field names.
2. Select suitable data types.
3. Identify the primary key.
4. Create the table in Microsoft Access.
5. Enter at least five records.

19. Home Assignment

Create a Microsoft Access database named:

`LibraryDB.accdb`

Create a table named:

`Books`

The table should contain at least seven fields and ten records.

Students must identify:

- Field names
- Appropriate data types
- Primary key

End of Lecture 01

Microsoft Access: Database Basics, Fields, Tables and Keys

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Academic Year: 2026